

**ASSESSMENT TOOL 3 – INCIDENT RESPONSE (MONITORING
& MITIGATION)**

Cour	CSA5802	
	CO5 – Precision (BL1, BL2,	A
Weightage:	30% of C	Tc
CO5 Sta	Develop Incident Response Plan, Ana RecommendMitigation Strategies and	
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Section	/ Batch: 2024	

- Develop an incident response plan based on the given security incident scenario and identified findings.
- Analyze the provided security monitoring dashboard, alerts, logs and indicators to determine incident status and affected assets.
- Recommend prioritized mitigation strategies that directly address the identified threats, vulnerabilities and incident impact.
- Provide clear technical justification for each major response and mitigation decision using evidence from the scenario.
- Present the response plan, dashboard analysis and recommendations clearly and professionally within the prescribed submission format.

INCIDENT RESPONSE Monitoring & Mitigation Q1

Q1–100

Q – Incident Response, Monitoring & Mitigation Scenario. A campus-network security dashboard reports that a student endpoint at 10.20.15.42 is generating repeated connection attempts against hosts in multiple internal VLANs. The device is assigned to the Student VLAN, but alerts show scanning activity toward Faculty and Server VLANs. The activity includes common TCP service ports such as 22, 80, 443, 445 and 3389. The IDS/IPS marks the pattern as high severity, while NAC reports a policy violation because the endpoint is attempting unauthorized east-west discovery.

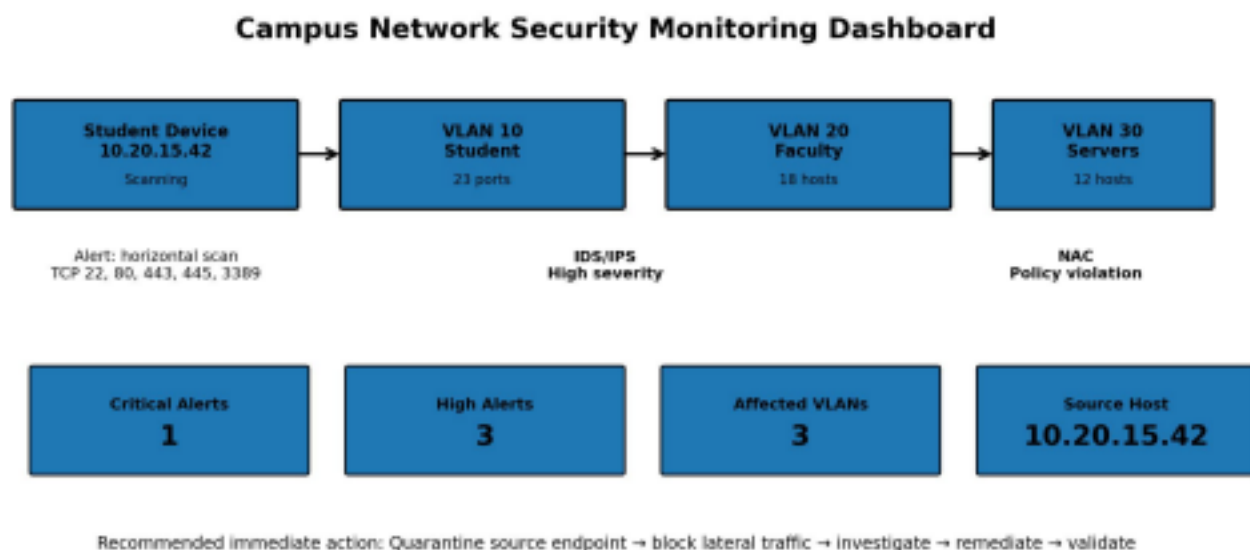


Figure 1. Simulated campus-network security monitoring dashboard.

1. Incident Identification and Status

- Incident type: Unauthorized internal network scanning / possible lateral-movement reconnaissance. • Primary source asset: Student endpoint 10.20.15.42.
- Affected security zones: Student VLAN, Faculty VLAN and Server VLAN.
- Severity: High, because the activity crosses trust boundaries and targets multiple internal assets. • Current status: Active and requires immediate containment; compromise is not yet confirmed. • Key indicators: repeated connection attempts, multi-VLAN targeting, policy violation, and IDS/IPShigh-severity alert.

2. Dashboard Analysis

Indicator	Observed	Meaning
Source	10.20.15.42	Student endpoint
Target VLANs	10, 20, 30	Traffic to multiple VLANs

Ports	22, 80, 443, 445, 3389	Discovery ofSSH/HTTP/HTTPS
IDS/IPS	High severity	Reconnaissance pattern
NAC	Policy violation	Endpoint posture/access failed
Impact	No confirmed compromise	Potential for lateral movement

3. Incident Response Plan

Incident Response Flow



Evidence preservation and documentation occur throughout every phase.

Figure 2. Incident response lifecycle for the campus scanning incident.

1. Preparation: Maintain incident contacts, NAC policies, VLAN/ACL diagrams, IDS/IPS rules, endpoint tooling, logging and evidence-retention procedures.

2. Detection & Triage: Correlate NAC, IDS/IPS, switch, firewall, DHCP and endpoint logs. Confirm the source MAC/user, scan rate, targets and first-seen time. Determine whether any successful connections followed the scan.

3. Containment: Immediately quarantine 10.20.15.42 using NAC. Block its inter-VLAN traffic with switch/firewall controls while preserving management access needed for investigation.

4. Investigation: Collect endpoint alerts, running processes, recent downloads, authentication events, DNS history, network connections and relevant packet/log evidence. Identify whether the scan was caused by malware, a vulnerable application, a security tool or intentional activity.

5. Eradication: Remove malicious or unauthorized software, patch vulnerable applications/OS components, disable compromised accounts or tokens, reset credentials when exposure is suspected, and rescan the endpoint.

6. Recovery: Restore the endpoint to normal network access only after security checks pass. Monitor it closely and validate that cross-VLAN scanning has stopped.

7. Lessons Learned: Document timeline, root cause, control gaps and false positives. Tune IDS/IPS signatures, NAC posture rules and inter-VLAN ACLs; update student acceptable-use and security guidance.

4. Recommended Mitigation Strategies

Control	Recommendation	Technical Justification	Priority
NAC	Quarantine the source endpoint, require compliant posture before reconnecting.	NAC provides identity/device access control and move a vulnerable endpoint to a restricted quarantine. This stops the source from disabling an entire VLAN.	P1 can't access a network without
VLAN Isolation	Keep Student, Faculty/Server services in separate VLANs and restrict inter-VLAN communication using privilege ACLs/firewall rules.	Segmentation limits lateral movement. A student device should not freely reach administrative or server lease services unless explicitly required.	P1

IDS/IPS	Use signatures/behavior analysis to detect horizontal scans and block repeated malicious patterns.	IDS provides visibility; IPS adds prevention by dropping traffic that matches high-confidence attack patterns. Alerts should be correlated with source IP/MAC/VLAN.	P1 and
Endpoint Remediation	Run EDR/antimalware patch the OS/apps, remove unauthorized review persistence, and reset credentials.	The endpoint is the source of the activity. Network controls alone may not remove the underlying cause; remediation reduces recurrence.	P1 of the
Logging/SIEM	Centralize NAC, firewall, switch, IDS/IPS, DHCP endpoint events.	Correlation supports accurate reconstruction helps detect repeat scanning and or successful lateral movement.	P2 and
Access Control	Apply least privileged deny-by-default rules sensitive server ports SMB/RDP/SSH.	Reduces the number of and services reachable from for student devices and such as blast radius of a compromised endpoint.	P2 limits the

5. Proposed Technical Architecture

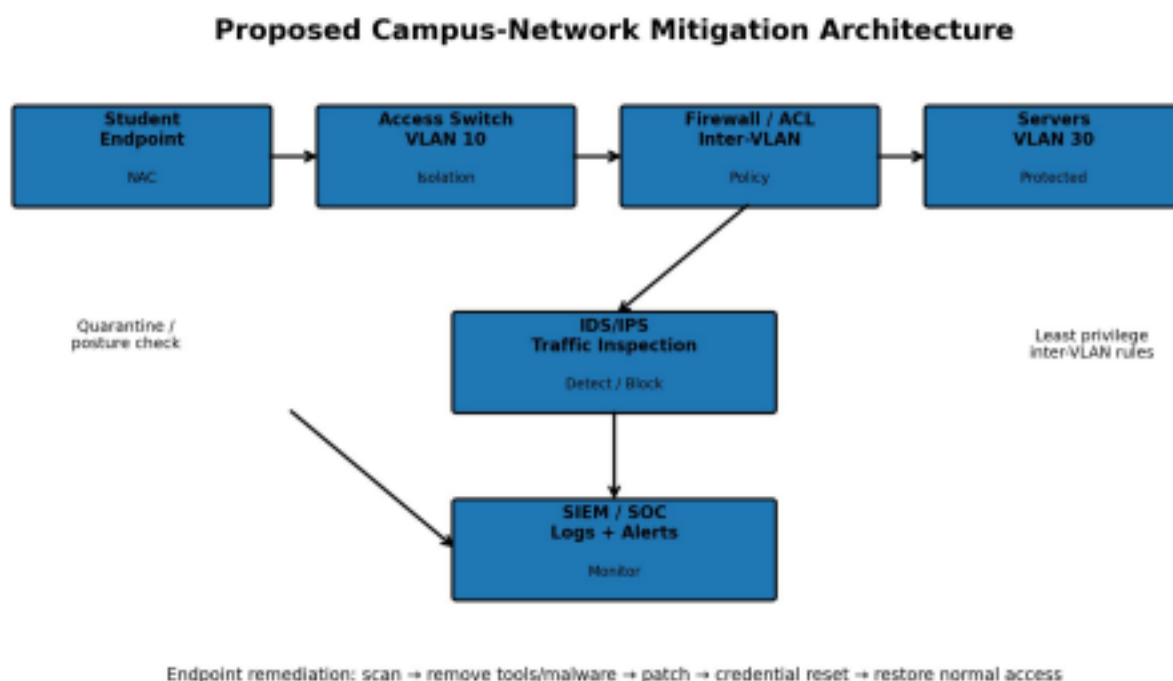


Figure 3. NAC, VLAN isolation, IDS/IPS and endpoint remediation architecture.

6. Technical Justification

- NAC is the first containment control because the incident originates from a known campus endpoint. It can associate

the device with its identity and posture and place it into a quarantine role. • VLAN isolation is essential because the dashboard shows cross-VLAN reconnaissance. Segmentation creates a security boundary and prevents unrestricted east-west traffic.

- IDS/IPS is required for continuous detection of scanning behavior. IDS gives visibility while IPS can automatically block high-confidence malicious traffic.
- Endpoint remediation is necessary because scanning may result from malware, unauthorized security tools, a vulnerable service, or intentional misuse. Removing the cause prevents the same endpoint from returning to the network and repeating the activity.
- Layered controls are stronger than any single control: NAC controls access, VLAN/ACL controls reachability, IDS/IPS detects and blocks suspicious traffic, and endpoint security removes the root cause.

7. Example Containment and Verification Checklist

- ☐ Identify user, device MAC address, switch port, DHCP lease and VLAN for 10.20.15.42. • ☐ Place the endpoint in NAC quarantine.
- ☐ Block unnecessary inter-VLAN traffic from the quarantine/student role. • ☐ Preserve relevant logs before changing configurations that could destroy evidence. • ☐ Check for successful connections after the scan and inspect targeted hosts. • ☐ Perform endpoint malware/EDR scan and vulnerability assessment.
- ☐ Patch systems and remove unauthorized software.
- ☐ Reset credentials/tokens if compromise is suspected.
- ☐ Re-test network behavior and confirm no scanning remains.
- ☐ Restore normal access only after validation.
- ☐ Record root cause, actions, evidence and lessons learned.

8. Expected Outcome

The incident should be considered contained when the source endpoint is quarantined, unauthorized cross-VLAN traffic is blocked, no successful lateral movement is identified, the endpoint passes security checks, and monitoring confirms that scanning has stopped. Normal access should be restored gradually and remains subject to least-privilege network policy.

9. Conclusion

The dashboard indicates a high-risk internal reconnaissance event originating from a student device. The recommended response combines rapid NAC quarantine with VLAN isolation, IDS/IPS monitoring and blocking, and complete endpoint remediation. This layered DevSecOps security approach reduces the probability of lateral movement, limits incident impact, preserves evidence, and improves continuous security monitoring across the campus network.